

Chapter 1 - Mathematical Models

1.1

Write the differential equation for a diffusion problem where the diffusing element decays at a rate proportional to the concentration.

Solution:

The diffusion equation in one dimension is

$$\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} + \gamma u = f$$

where u is the concentration of the element and f is the source. The model for this problem is

$$\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} = -\gamma u$$

Writing this out in three dimensions, it instead becomes

$$\frac{\partial u}{\partial t} - D \Delta u = -\gamma u$$

where $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

□

1.3

The temperature distribution in a thin isolated rod is $\frac{T_0}{1+x^2}$ at a specific moment in time. Calculate the heat flux. In what way is it directed?

Solution:

The heat flux is

$$\mathbf{j} = -\lambda \nabla T = \frac{\lambda 2x T_0}{(1+x^2)^2} \hat{x}$$

The direction is $\hat{x}$

□

1.4

Determine a differential equation for heat conduction in a thin rod, not perfectly isolated. The heat loss at the point x is proportional to the difference between the rod temperature and its surrounding temperature.

Solution:

$$\frac{1}{a} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = -\frac{c}{\lambda} (u - T_0)$$

□

1.5

Determine a differential equation for transversal vertical oscillations of a long horizontal string

(a) under the influence of gravity,

(b) if the surrounding medium additionally exerts a resistance proportional to the velocity.

Solution:

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = f$$

(a) Here, f is simply the gravitational force $F = mg$

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = -\frac{mg}{\rho_l}$$

where ρ_l is the length density.

(b) Add a proportionality factor of $\frac{\partial u}{\partial x}$ which is the velocity

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = -\frac{mg}{\rho_l} - a \frac{\partial u}{\partial x}$$

□

1.8

Let $u(x, y)$ be the temperature in a plate Ω .

(a) How do you formulate a mathematical heat conduction problem with homogenous Neumann boundary conditions?

(b) What physical phenomenon gives rise to homogenous Neumann boundary conditions? Motivate.

In the figures below, the solutions to the problem is shown. Along some boundaries there are homogenous Neumann conditions, and along some there are homogenous Dirichlet conditions.

(c) Left figure shows the vector field, ∇u . Along which parts of the boundary is there homogenous Neumann conditions?

(d) Right figure shows the isotherms ($u = \text{constant}$). Along which parts of the boundary is there homogenous Neumann conditions?

Solution:

(a)

$$\frac{1}{a} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f$$

$$\frac{\partial u}{\partial x} = 0, \text{ on some boundary}$$

$$\frac{\partial u}{\partial y} = 0, \text{ on some boundary}$$

(b) Heat flux on boundary

(c) Top, top right, bottom left.

(d) Left, bottom and top right. □

1.10**Solution:**

□

1.11**Solution:**

□

1.12

A wide and tall wall with thickness L starts with temperature T_0 , same as surroundings. At time $t = 0$ the temperature on one side changes to T_1 . At the boundaries on the wall the coefficients are α_0 and α_1 . Determine the differential equation, initial values and boundary conditions.

Solution:

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0$$

$$u(L, t) = T_0$$

$$\frac{\partial u(0, t)}{\partial x} = \alpha_0(u(0, t) - T_1)$$

$$\frac{\partial u(L, t)}{\partial x} = -\alpha_1(u(L, t) - T_0)$$

□

1.13

Interpret the following in physical meaning for heat conduction.

(a) $u(0, t) = 0$

(b) $\frac{\partial u(0, t)}{\partial x} = 0$

(c) $\frac{\partial u(0, t)}{\partial x} - hu(0, t) = 0, \frac{\partial u(L, t)}{\partial x} + hu(L, t) = 0$

Solution:

- (a) Temperature at $x = 0$ always 0.
 - (b) No heat flux at $x = 0$, isolated.
 - (c) Heat flux on both boundaries, proportional to the difference in temperature between the medium and the surroundings. $\square$
-

1.14**Solution:** $\square$

1.15**Solution:** $\square$

1.18**Solution:** $\square$

1.21**Solution:** $\square$

1.22

Solution:

☐

1.23

Solution:

☐

1.26

Solution:

☐

Chapter 3.1–3.3 - Fourier's Method I

3.2

Determine the coefficients in the following expansions

(a)

$$x = \sum_0^{\infty} \alpha_k \cos(k\pi x), \quad 0 < x < 1$$

(b)

$$x = \sum_1^{\infty} \alpha_k \sin(k\pi x), \quad 0 < x < 1$$

Solution:

(a)

$$\alpha_k = 2 \int_0^1 x \cos(k\pi x) \, dx = 2 \underbrace{\left[\frac{-x \sin(k\pi x)}{k\pi} \right]_0^1}_{=0} + 2 \int_0^1 \frac{\sin(k\pi x)}{k\pi} \, dx =$$

$$= \frac{2}{k\pi} \int_0^1 \sin(k\pi x) \, dx = \frac{2}{k\pi} \left[\frac{\cos(k\pi x)}{k\pi} \right]_0^1 = \frac{2 \cos(k\pi)}{k^2 \pi^2} - \frac{2}{k^2 \pi^2} =$$

$$= \frac{2}{k^2 \pi^2} (\cos(k\pi) - 1) = \frac{2}{k^2 \pi^2} ((-1)^k - 1)$$

$$\alpha_0 = \int_0^1 x \, dx = \frac{1}{2}$$

$$\alpha_k = \frac{2}{k^2 \pi^2} ((-1)^k - 1)$$

↓

$$x = \frac{1}{2} + \frac{2}{\pi^2} \sum_0^{\infty} \frac{(-1)^k - 1}{k^2} \cos(k\pi x)$$

(b)

$$\begin{aligned}
\beta_k &= 2 \int_0^1 x \sin(k\pi x) \, dx = 2 \left[\frac{x \cos(k\pi x)}{k\pi} \right]_0^1 - \underbrace{2 \int_0^1 \frac{\cos(k\pi x)}{k\pi} \, dx}_{=0} = \\
&= \frac{2 \cos(k\pi)}{k\pi} \\
\beta_0 &= \int_0^1 x \sin(0\pi x) \, dx = 0 \\
&\quad \downarrow \\
x &= \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \sin(k\pi x)
\end{aligned}$$

□

3A.5

Solve the heat problem

$$\begin{aligned}
\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < 1, t > 0 \\
u(0, t) = u(1, t) &= 0, & t > 0 \\
u(x, 0) &= \sin(\pi x) + 2 \sin(3\pi x), & 0 < x < 1
\end{aligned}$$

Solution:

We have homogenous Dirichlet conditions, which should give our solution in the form of a sinus series. We can therefore express our solution in an infinite sum

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin\left(\frac{k\pi x}{L}\right)$$

with $L = 1$. Using this expression we can differentiate termwise and put into the differential equation, resulting in

$$\begin{aligned}
& \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0 \\
& \downarrow \\
& \sum_{k=1}^{\infty} \underbrace{(u'_k(t) + k^2 \pi^2 u_{k(t)})}_{=0} \sin(k\pi x) = 0 \\
& \downarrow \\
& u'_k(t) + k^2 \pi^2 u_{k(t)} = 0 \\
& \downarrow \\
& u_k(t) = c_k e^{-k^2 \pi^2 t}
\end{aligned}$$

This gives our function $u(x, t)$ as

$$u(x, t) = \sum_{k=1}^{\infty} c_k e^{-k^2 \pi^2 t} \sin(k\pi x)$$

and using our initial conditions for $u(x, 0)$, we can determine the coefficients c_k :

$$\begin{aligned}
u(x, 0) &= \sum_{k=1}^{\infty} c_k \sin(k\pi x) = \sin(\pi x) + 2 \sin(3\pi x) \\
&\downarrow \\
c_k &= \begin{cases} 1, & k = 1 \\ 2, & k = 3 \\ 0, & \text{otherwise} \end{cases}
\end{aligned}$$

This gives the final solution

$$u(x, t) = e^{-\pi^2 t} \sin(k\pi x) + 2e^{-9\pi^2 t} \sin(3\pi x)$$

□

3A.7

A thin rod with closed ends has length $L = 1$. In the rod is air mixed with gas. Initially, the gas concentration is q in the left half of the rod, and 0 in the right half of the rod. Diffusion constant is set to $D = 1$. There is no source inside the rod. Write down a mathematical model and solve the problem.

Solution:

Using the diffusion equation

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < 1, t > 0 \\ \frac{\partial u(0, t)}{\partial x} &= \frac{\partial u(1, t)}{\partial x} = 0, & t > 0 \\ u(x, 0) &= \begin{cases} q, & 0 < x < \frac{1}{2} \\ 0, & \frac{1}{2} < x < 1 \end{cases}\end{aligned}$$

We have homogenous Neumann conditions, which will give the solution on the form of a cosine series expansion.

$$u(x, t) = \sum_{k=0}^{\infty} u_k(t) \cos(k\pi x)$$

Putting this into the differential equation and differentiating termwise gives

$$\begin{aligned}\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0 \\ \downarrow \\ \sum_{k=0}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \cos(k\pi x) &= 0 \\ \downarrow \\ u'_{k(t)} + k^2 \pi^2 u_k(t) &= 0 \\ \downarrow \\ u_k(t) &= c_k e^{-k^2 \pi^2 t}\end{aligned}$$

This gives the solution on the form

$$u(x, t) = \sum_{k=0}^{\infty} c_k e^{-k^2 \pi^2 t} \cos(k\pi x)$$

and we only need to determine the coefficients c_k , which is done by the initial value $u(x, 0)$.

We want to expand the function $u(x, 0)$ as a cosine series

$$\begin{aligned}u(x, 0) &= \sum_{k=0}^{\infty} c_k \cos(k\pi x) \\ \downarrow \\ c_k &= 2 \int_0^{\frac{1}{2}} q \cos(k\pi x) \, dx = 2 \left[\frac{q \sin(k\pi x)}{k\pi} \right]_0^{\frac{1}{2}} = \frac{2q \sin(\frac{k\pi}{2})}{k\pi} \\ c_0 &= \int_0^{\frac{1}{2}} q \, dx = \frac{q}{2}\end{aligned}$$

Final solution is then

$$u(x, t) = \frac{q}{2} + \frac{2q}{\pi} \sum_{k=1}^{\infty} \frac{\sin\left(\frac{k\pi}{2}\right)}{k} e^{-k^2 \pi^2 t} \cos(\pi k x)$$

□

3A.8

Solve the heat problem

$$\begin{aligned} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad 0 < x < 1, t > 0 \\ u(x, 0) &= x, \quad 0 < x < 1 \end{aligned}$$

with the boundary conditions

(a)

$$u(0, t) = u(1, t) = 0, \quad t > 0$$

(b)

$$\frac{\partial u(0, t)}{\partial x} = \frac{\partial u(1, t)}{\partial x} = 0, \quad t > 0$$

Solution:

(a) This problem has homogenous Dirichlet conditions, which will give the solution on sinus form.

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin(k\pi x)$$

Putting this into the differential equation and differentiating termwise gives

$$\begin{aligned} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0 \\ \downarrow \\ \sum_{k=1}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \sin(k\pi x) &= 0 \\ \downarrow \\ u'_k(t) + k^2 \pi^2 u_k(t) &= 0 \\ \downarrow \\ u_k(t) &= c_k e^{-k^2 \pi^2 t} \end{aligned}$$

This gives

$$u(x, t) = \sum_{k=1}^{\infty} c_k e^{-k^2 \pi^2 t} \sin(k\pi x)$$

and

$$u(x, 0) = x = \sum_{k=1}^{\infty} c_k \sin(k\pi x)$$

from which we can determine the coefficients c_k .

$$c_k = 2 \int_0^1 x \sin(k\pi x) dx = 2 \left[\frac{-x \cos(k\pi x)}{k\pi} \right]_0^1 + \underbrace{2 \int_0^1 \frac{\cos(k\pi x)}{k\pi} dx}_{=0}$$

$$c_k = \frac{-2 \cos(k\pi)}{k\pi} = \frac{2}{k\pi} (-1)^{k+1}$$

$$c_0 = \int_0^1 x \sin(k\pi 0) dx = 0$$

Final solution is

$$u(x, t) = \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} e^{-k^2 \pi^2 t} \sin(k\pi x)$$

(b) This problem has homogenous Neumann conditions, which will give the solution on cosine form.

$$u(x, t) = \sum_{k=0}^{\infty} u_k(t) \cos(k\pi x)$$

Putting into differential equation

$$\sum_{k=0}^{\infty} (u'_k(t) + k^2 \pi^2 u_k(t)) \cos(k\pi x) = 0$$

↓

$$u'_k(t) + k^2 \pi^2 u_k(t) = 0$$

↓

$$u_k(t) = c_k e^{-k^2 \pi^2 t}$$

$$u(x, t) = \sum_{k=0}^{\infty} c_k e^{-k^2 \pi^2 t} \cos(k\pi x)$$

$$u(x, 0) = x = \sum_{k=0}^{\infty} c_k \cos(k\pi x)$$

From this we can determine the coefficients c_k

$$\begin{aligned}
 c_k &= 2 \int_0^1 x \cos(k\pi x) \, dx = -2 \int_0^1 \frac{\sin(k\pi x)}{k\pi} \, dx = \\
 &= 2 \left[\frac{\cos(k\pi x)}{k^2 \pi^2} \right]_0^1 = \frac{2 \cos(k\pi)}{k^2 \pi^2} - \frac{2}{k^2 \pi^2} = \frac{2}{k^2 \pi^2} ((-1)^k - 1) \\
 c_0 &= \int_0^1 x \cos(k\pi 0) \, dx = \frac{1}{2}
 \end{aligned}$$

This gives our final solution as

$$u(x, t) = \frac{1}{2} + \frac{2}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^k - 1}{k^2} e^{-k^2 \pi^2 t} \cos(k\pi x)$$

□

3A.10

Solution:

□

3A.12

Solve the oscillation problem

$$\begin{aligned}
 \frac{\partial^2 u}{\partial t^2} + 2 \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} &= 0, \quad 0 < x < 1, t > 0 \\
 u(0, t) = u(1, t) &= 0, \quad t > 0 \\
 u(x, 0) &= 1, \quad 0 < x < 1 \\
 \frac{\partial u(x, 0)}{\partial t} &= 0, \quad 0 < x < 1
 \end{aligned}$$

Solution:

We have homogenous Dirichlet conditions which will give the solution on sinus form

$$u(x, t) = \sum_{k=1}^{\infty} u_k(t) \sin(k\pi x)$$

Insert into differential equation and differentiate termwise

$$\sum_{k=1}^{\infty} (u_k''(t) + 2u_k'(t) + k^2\pi^2 u_k(t)) \sin(k\pi x) = 0$$

↓

$$u_k''(t) + 2u_k'(t) + k^2\pi^2 u_k(t)$$

Solving the characteristic equation with $u_k = re^t$

$$r^2 + 2r + k^2\pi^2 = 0$$

↓

$$r = -1 + \sqrt{1 - k^2\pi^2} = -1 \pm i\sqrt{k^2\pi^2 - 1} = -1 \pm i\omega_k$$

↓

$$u_k(t) = e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t))$$

Gives the solution as

$$u(x, t) = \sum_{k=1}^{\infty} e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t)) \sin(k\pi x)$$

$$u(x, 0) = 1 = \sum_{k=1}^{\infty} a_k \sin(k\pi x)$$

↓

$$a_k = 2 \int_0^1 \sin(k\pi x) dx = 2 \left[\frac{-\cos(k\pi x)}{k\pi} \right]_0^1 = \frac{-2 \cos(k\pi)}{k\pi} + \frac{2}{k\pi} \rightarrow a_k = \frac{2}{k\pi} (1 - (-1)^k)$$

$$\frac{\partial u(x, 0)}{\partial t} = 0 =$$

$$= \sum_{k=1}^{\infty} (-e^{-t}(a_k \cos(\omega_k t) + b_k \sin(\omega_k t)) + e^{-t}(-a_k \omega_k \sin(\omega_k t) + b_k \omega_k \cos(\omega_k t))) \sin(k\pi x) =$$

$$= \sum_{k=1}^{\infty} (-a_k + \omega_k b_k) \sin(k\pi x) = 0$$

↓

$$-a_k + \omega_k b_k = 0 \rightarrow b_k = \frac{a_k}{\omega_k} = \frac{2}{\omega_k k\pi} (1 - (-1)^k)$$

Full solution:

$$u(x, t) = \frac{2}{\pi} \sum_{k=1}^{\infty} e^{-t} \frac{1 - (-1)^k}{k} \left(\cos(\omega_k t) + \frac{\sin(\omega_k t)}{\omega_k} \right) \sin(k\pi x)$$

□

3A.13

Solution:

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3A.14

Solution:

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3A.15

Solution:

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3A.17

Solution:

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3A.19

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3A.21

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3A.22

Solution:

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3A.24a

Solution:

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3A.24bc

Solution:

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3A.25

Solution:

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Chapter 3.4–3.5 - Fourier's Method I

3B.29

Solution:

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3B.30

Solution:

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3B.32a

Solution:

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3B.35

Solution:

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3B.39

Solution:

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3B.41

Solution:

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3B.42

Solution:

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3B.43+60

Solution:

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3B.45a

Solution:

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3B.46

Solution:

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3B.49a

Solution:

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3B.51

Solution:

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3B.55

Solution:

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3B.62

Solution:

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3B.66

Solution:

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3B.69

Solution:

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Chapter 4 - Fourier's Method II

4.1

Solution:

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4.3

Solution:

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4.20

Solution:

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Chapter 5 - Green's Functions

5.1

Solution:

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5.2

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5.4

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5.9

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5.27

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5.28

Solution:

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Chapter 7 - Radiating Waves

7.1

Solution:

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7.2

Solution:

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7.3

Solution:

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7.4

Solution:

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7.6

Solution:

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7.8ab

Solution:

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7.8c

Solution:

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7.9

Solution:

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7.10

Solution:

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7.18

Solution:

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Distributions

D.1

Solution:

$$\begin{aligned}\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} &= -\frac{m_0 g}{\rho} \delta(x-a), \quad 0 < x < L, t > 0 \\ u(0, t) &= u(L, t) = 0, \quad t > 0 \\ u(x, t) &= g(x), \quad \frac{\partial u(x, 0)}{\partial t} = h(x), \quad 0 < x < L\end{aligned}$$

□

D.4

- (a) $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$
 (b) $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$

Solution:

- (a) Here we use $g(x)\delta(x) = g(0)\delta(x)$, and start by differentiating the expression $(g(x)\delta(x))$:

$$\begin{aligned}(g(x)\delta(x))' &= g'(x)\delta(x) + g(x)\delta'(x) \\ &\downarrow \\ g(x)\delta'(x) &= (g(x)\delta(x))' - g'(x)\delta(x) = \\ &= (g(0)\delta(x))' - g'(0)\delta(x) = g(0)\delta'(x) - g'(0)\delta(x)\end{aligned}$$

- (b) Using the result from (a) and differentiating $(g(x)\delta'(x))$:

$$\begin{aligned}
 (g(x)\delta'(x))' &= g'(x)\delta'(x) + g(x)\delta''(x) \\
 &\downarrow \\
 g(x)\delta''(x) &= (g(x)\delta'(x))' - g'(x)\delta'(x) = \\
 &= (g(0)\delta'(x) - g'(0)\delta(x))' - g'(x)\delta'(x) = \\
 &= g(0)\delta''(x) - g'(0)\delta'(x) - g'(0)\delta'(x) + g''(0)\delta(x) = \\
 &= g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)
 \end{aligned}$$

□

D.5

- (a) $e^x \delta_1$
- (b) $x\delta'$
- (c) $\cos(x)\delta'$
- (d) $x^2\delta''$
- (e) $\int g(t)\delta(t-1) dt$

Solution:

(a)

$$e^x \delta_1 = e^x \delta_1 = e \delta_1$$

(b) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$

$$x\delta' = -\delta$$

(c) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$:

$$\cos(x)\delta' = \cos(0)\delta' + \sin(0)\delta = \delta'$$

(d) Using $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$:

$$x^2\delta'' = 2\delta$$

(e) Using $\int g(x)\delta(x-a) dx = g(a)$:

$$\int g(t)\delta(t-1)dt = g(1)$$

□

D.6

(a) $f(x) = e^{-x}\theta(x)$

(b) $f(x) = e^{-|x|}$

Solution:

(a) Using $\theta'(x) = \delta(x)$

$$f'(x) = (e^{-x}\theta(x))' = -e^{-x}\theta(x) + e^{-x}\delta(x) = \delta(x) - e^{-x}\theta(x)$$

$$f''(x) = (\delta(x) - e^{-x}\theta(x))' = \delta'(x) + e^{-x}\theta(x) - e^{-x}\delta(x) = \delta'(x) - \delta(x) + e^{-x}\theta(x)$$

(b) Expressing $|x|$ in distributions:

$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases} = x\theta(x) - x(1 - \theta(x)) = 2x\theta(x) - x$$

$$f'(x) = (-2\theta(x) - 2x\delta(x) + 1)e^{-|x|} = (1 - 2\theta(x))e^{-|x|}$$

$$\begin{aligned} f''(x) &= (e^{-|x|} - 2e^{-|x|}\theta(x))' = (1 - 2\theta(x))e^{-|x|} + 2(1 - 2\theta(x))e^{-|x|}\theta(x) - 2e^{-|x|}\delta(x) = \\ &= e^{-|x|}[(1 - 2\theta(x)) + 2\theta(x)(1 - 2\theta(x))] - 2\delta(x) = e^{-|x|}(1 - 2\theta(x))^2 - 2\delta(x) = \\ &= [(1 - 2\theta(x))^2 = 1 \text{ for all } x \neq 0] = e^{-|x|} - 2\delta(x) \end{aligned}$$

□

D.8

(b) Determine all primitive distributions for $x\theta(x-1)$

Solution:

(a)

$$(x^2\theta(x-1))' = 2x\theta(x-1) + x^2\theta'(x-1) = 2x\theta(x-1) + \delta(x-1)$$

(b) Try to compensate for the $\delta(x-1)$ in (a) by $x^2 \rightarrow x^2 - 1$:

$$((x^2 - 1)\theta(x-1))' = 2x\theta(x-1) + (x^2 - 1)\delta(x-1) = 2x\theta(x-1)$$

Now compensate with a factor 2 and we get the solution

$$\left(\frac{x^2 - 1}{2}\right)\theta(x-1) + C$$

□

D.9

Solution:

$$\frac{\partial^2 u}{\partial t^2} - S \frac{\partial^2 u}{\partial x^2} = F\delta(x-a)$$

$$u(0, t) = u(1, t) = 0$$

Since we are looking for the stationary solution, $\frac{\partial^2 u}{\partial t^2} = 0$ and we have

$$-\frac{\partial^2 u}{\partial x^2} = \frac{F}{S}\delta(x-a)$$

Integrating both sides twice gives

$$-u(x, t) = \frac{F}{S}(x-a)\theta(x-a) + Ax + B$$

Now check the boundary conditions for this expression:

$$u(0, t) = \underbrace{\frac{F}{S}(-a)\theta(-a)}_{=0 \text{ since } a>0} + B = 0 \rightarrow B = 0$$

$$u(1, t) = \frac{F}{S}(a-1)\theta(1-a) + A = 0 \rightarrow A = \frac{F}{S}(1-a)$$

$$u(x, t) = \frac{F}{S}((x(1-a) + (a-x)\theta(x-a)))$$

□

D.10**Solution:**

□

D.12

$$u'(x) + (2x + 1)u(x) = \delta'(x)$$

Solution:

Integrating factor for LHS is

$$e^{x^2+x}$$

Multiply this into the equation and integrate LHS, then using the expansion for $f(x)\delta'(x)$:

$$\begin{aligned} \left(u(x)e^{x^2+x}\right)' &= e^{x^2+x}\delta'(x) = e^0\delta'(x) - (2x+1)_{x=0}\delta(x) = \\ &= \left(u(x)e^{x^2+x}\right)' = \delta'(x) - \delta(x) \end{aligned}$$

Now integrating both sides gives:

$$u(x)e^{x^2+x} = \delta(x) - \theta(x) + C$$

$$\downarrow$$

$$u(x) = e^{-(x^2+x)}(\delta(x) - \theta(x) + C) = \delta(x) + e^{-x^2-x}(C - \theta(x))$$

□

D.15

- (a) $xU = 0$
- (b) $(x-1)U = 0$
- (c) $(e^x - 2)U = 0$
- (d) $(x^2 - 1)U = 0$
- (e) $\sin(x)U = 0$
- (f) $x^3U = 0$

Solution:

(a)

$$U = \delta(x) \rightarrow x\delta(x) = 0 \rightarrow U = c\delta(x), \quad c = \text{constant}$$

(b) Same as in (a) but $\delta(x-1)$

$$U = c\delta(x-1)$$

(c) We can try to set U as $\delta(x-a)$ to make $(e^x - 2) = 0 \rightarrow e^x = 2 \rightarrow x = \ln(2)$

$$U = c\delta(x - \ln(2))$$

(d) Same idea as above, but here we have two roots, $x = 1, x = -1$

$$U = c_1\delta(x-1) + c_2\delta(x+1)$$

(e) Can use $U = \delta(x-a)$ again to make $\sin(x) = 0$

$$U = c_k\delta(x - k\pi) = \sum_{k=-\infty}^{\infty} c_k\delta(x - k\pi)$$

(f) Triple root at $x = 0$, use the relation $x^k U = 0 \rightarrow U = \sum_{k=0}^{n-1} c_k \delta^{(k)}(x)$

$$U = c_0\delta(x) + c_1\delta'(x) + c_2\delta''(x)$$

□

D.17(a) $\theta(-2x+1)$ (b) $\delta(2x)$ (c) $\delta(-2x+1)$ (d) $\delta'(2x)$ **Solution:**

(a)

$$\theta(-2x+1) = \begin{cases} 1, & -2x+1 > 0 \rightarrow x < \frac{1}{2} \\ 0, & -2x+1 < 0 \rightarrow x > \frac{1}{2} \end{cases}$$

$$\downarrow$$

$$\theta(-2x+1) = \theta\left(-2\left(x - \frac{1}{2}\right)\right) = 1 - \theta\left(x - \frac{1}{2}\right) = 1 - \theta_{\frac{1}{2}}(x)$$

(b) Scaling rule gives

$$\delta(2x) = \frac{1}{2}\delta(x)$$

(c)

$$\delta(-x) = \delta(x), \delta(ax) = \frac{1}{|a|}\delta(x)$$

$$\downarrow$$

$$\delta(-2x+1) = \delta\left(-2\left(x - \frac{1}{2}\right)\right) = \frac{1}{2}\delta\left(x - \frac{1}{2}\right) = \frac{1}{2}\delta_{\frac{1}{2}}(x)$$

(d)

$$\delta'(ax) = \frac{1}{a|a|}\delta'(x)$$

$$\downarrow$$

$$\delta'(2x) = \frac{1}{4}\delta'(x)$$

□

D.18

(a) $\delta * e^{-x^2}$

(b) $\delta' * (e^{-x}\theta(x))$

(c) $\delta_1 * \sin(2x)$

Solution:

Using the definition of the convolution

$$\varphi * U = \int \varphi(x-y)U(y) \, \mathrm{d}$$

(a)

$$\int e^{-(x-y)^2} \delta(y) \, dy = e^{-x^2}$$

(b)

$$\begin{aligned} \int e^{-(x-y)} \theta(x-y) \delta'(y) \, dy &= [e^{y-x} \theta(x-y) \delta(y)]_{-\infty}^{\infty} - \int (e^{y-x} \theta(x-y))' \delta(y) \, dy = \\ &= (e^{-x} \theta(x))' = -e^{-x} \theta(x) + e^{-x} \delta(x) = \delta(x) - e^{-x} \theta(x) \end{aligned}$$

(c)

$$\int \sin(2(x-y)) \delta(y-1) \, dy = \sin(2x-2)$$

□

D.22

Solution:

Definition of the cosine series expansion for distributions

$$\begin{aligned} U &= \frac{c_0}{2} + \sum_{k=1}^{\infty} c_k \cos\left(\frac{k\pi x}{L}\right), \quad 0 < x < L \\ c_k &= \frac{2}{L} \int_0^L U(x) \cos\left(\frac{k\pi x}{L}\right) \, dx \end{aligned}$$

We will also use the expansion of $f(x)\delta'(x) = f(0)\delta'(x) - f'(0)\delta(x)$.

This gives ($k \neq 0$)

$$\begin{aligned} c_k &= \frac{2}{\pi} \int_0^{\pi} \delta'\left(x - \frac{\pi}{2}\right) \cos(kx) \, dx = \frac{2}{\pi} \int_0^{\pi} \cos\left(k\frac{\pi}{2}\right) \delta'\left(x - \frac{\pi}{2}\right) + k \sin\left(k\frac{\pi}{2}\right) \delta\left(x - \frac{\pi}{2}\right) \, dx = \\ &= \frac{2}{\pi} \int_0^{\pi} k \sin\left(\frac{k\pi}{2}\right) \delta\left(x - \frac{\pi}{2}\right) \, dx = \frac{2}{\pi} k \sin\left(\frac{k\pi}{2}\right) \end{aligned}$$

and for $k = 0$:

$$c_0 = \frac{2}{\pi} \int_0^{\pi} \delta' \left(x - \frac{\pi}{2} \right) dx = \frac{2}{\pi} \left[\delta \left(x - \frac{\pi}{2} \right) \right]_0^{\pi} = 0$$

$$\delta' \left(x - \frac{\pi}{2} \right) = \frac{2}{\pi} \sum_{k=1}^{\infty} k \sin \left(\frac{k\pi}{2} \right) \cos(kx)$$

□

D.23

Solution:

Whut □

D.24

Solution:

Diffusion,

$$\begin{aligned} \frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < L, t > 0 \\ \frac{\partial u(0, t)}{\partial x} &= \frac{\partial u(L, t)}{\partial x} = 0, & t > 0 \\ u(x, 0) &= M \delta_{\frac{L}{2}}, & 0 < x < L \end{aligned}$$

We have homogenous Neumann boundary conditions, which should give us a solution in the form of a cosine series expansion.

$$u(x, t) = \sum u_k(t) \cos \left(\frac{k\pi x}{L} \right)$$

Differentiating termwise and putting into the differential equation gives

$$\sum (u'_k(t) + \alpha u_k(t)) \cos\left(\frac{k\pi x}{L}\right) = 0, \quad \alpha = \frac{Dk^2\pi^2}{L^2}$$

$$\downarrow$$

$$u'_k(t) + \alpha u_k(t) = 0$$

$$\downarrow$$

$$u_k(t) = c_k e^{-\alpha t}$$

We can also expand $u(x, 0) = M\delta_{\frac{L}{2}}$ in a cosine series, to get the coefficients, $u_k(0) = c_k$

$$u(x, 0) = M\delta_{\frac{L}{2}} = \sum a_k \cos\left(\frac{k\pi x}{L}\right)$$

$$c_0 = \frac{1}{L} \int_0^L M\delta_{\frac{L}{2}} dx = \frac{M}{L}$$

$$c_k = \frac{2}{L} \int_0^L M\delta_{\frac{L}{2}} \cos\left(\frac{k\pi x}{L}\right) dx = \frac{2M}{L} \cos\left(\frac{k\pi}{2}\right) = , \quad k = 1, 2, 3, \dots$$

This gives the full solution

$$u(x, t) = \frac{M}{L} \left(1 + 2 \sum_{k=1}^{\infty} \cos\left(\frac{k\pi}{2}\right) e^{-\alpha t} \cos\left(\frac{k\pi x}{L}\right) \right)$$

□

D.25

Solution:

□

D.27

Solve the three-dimensional potential problem, with $r = \sqrt{x^2 + y^2 + z^2}$

$$\Delta u = \delta(r - 2), \quad 1 < r < 3$$

$$u = 1, \quad r = 1$$

$$u = 3, \quad r = 3$$

Solution:

Symmetric in r so the Laplacian becomes

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = \delta(r-2)$$

$$\left(r^2 \frac{\partial u}{\partial r} \right)' = r^2 \delta(r-2) = 4\delta(r-2)$$

↓

$$r^2 \frac{\partial u}{\partial r} = 4\theta(r-2) + A$$

$$\frac{\partial u}{\partial r} = \frac{4}{r^2} \theta(r-2) + \frac{A}{r^2}$$

$$u(r) = \left(2 - \frac{4}{r} \right) \theta(r-2) - \frac{A}{r} + B$$

$$u(1) = 1 = -\theta(-1) - A + B = 0 \rightarrow -A + B = 1$$

$$u(3) = 3 = \left(\frac{2}{3} \right) \theta(1) - \frac{A}{3} + B = \frac{2}{3} + \frac{2}{3}A = 3$$

↓

$$\begin{cases} -A + B = 1 \\ \frac{2}{3} - \frac{A}{3} + B = 3 \end{cases} \rightarrow \frac{2}{3} + \frac{2}{3}A = 2 \rightarrow A = 2, B = 3$$

↓

$$u(r) = \left(2 - \frac{4}{r} \right) \theta(r-2) - \frac{2}{r} + 3$$

□

D.29**Solution:**

$$-\Delta u = -u + \delta(x-a)\delta(y-b), \quad 0 < x < 2, 0 < y < 1$$

$$-\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = -u + \delta_{a,b}$$

□

D.30

- (a) $\delta(t-1)$
- (b) $\delta'(t+2)$
- (c) 2
- (d) $\sin(2t)$
- (e) $(t+1)^2$
- (f) $e^t\theta(t)$

Solution:

(a) The Fourier transform of $\delta(x)$ is $\mathcal{F}[\delta(x)] = 1$ and the translation rule is $\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0} \mathcal{F}[f(t)]$

$$\mathcal{F}[\delta(t-1)] = e^{-i\omega}$$

(b) The derivative rule is $\mathcal{F}[f'(t)] = i\omega \mathcal{F}[f(t)]$. Using this and the translation rule

$$\mathcal{F}[\delta'(t+2)] = i\omega e^{2i\omega}$$

(c) Fourier transform of a constant is $2\pi\delta(\omega)$

$$\mathcal{F}[2] = 4\pi\delta(\omega)$$

(d)

$$\sin(2t) = \frac{e^{2it} - e^{-2it}}{2i}$$

We have $\mathcal{F}[e^{i\omega_0 t} f(t)] = \mathcal{F}[f(\omega - \omega_0)]$

$$\mathcal{F}\left[\frac{e^{2it}}{2i} \cdot 1 - \frac{e^{-2it}}{2i} \cdot 1\right] = \frac{\pi}{i}(\delta(\omega - 2) - \delta(\omega + 2))$$

(e) $\mathcal{F}[tf(t)] = i \frac{d}{d\omega} \mathcal{F}[f(t)]$

$$\begin{aligned} \mathcal{F}[(t+1)^2] &= \mathcal{F}[t^2 + 2t + 1] = i^2 \frac{d^2}{d\omega^2} 2\pi\delta(\omega) + i \frac{d}{d\omega} 4\pi\delta(\omega) + 2\pi\delta(\omega) = \\ &= -2\pi\delta''(\omega) + 4\pi i\delta'(\omega) + 2\pi\delta(\omega) \end{aligned}$$

(f) This distribution is not tempered, and thus cannot be Fourier transformed. $\square$

D.31

- (a) $\delta(\omega-1)$
- (b) $\delta'(\omega+2)$

(c) 2

(d) $\sin(2\omega)$ (e) $(\omega + 1)^2$ **Solution:**

Going backwards on the identities gives

(a)

$$\begin{aligned}
\mathcal{F}[1] &= 2\pi\delta(\omega) \\
\mathcal{F}[e^{i\omega_0 t} \cdot 1] &= 2\pi\delta(\omega - \omega_0) \\
&\downarrow \\
\mathcal{F}^{-1}[\delta(\omega - 1)] &= \frac{e^{it}}{2\pi}
\end{aligned}$$

(b)

$$\begin{aligned}
\mathcal{F}[t] &= 2\pi i\delta'(\omega) \\
\mathcal{F}[e^{i\omega_0 t} t] &= 2\pi i\delta'(\omega - \omega_0) \\
&\downarrow \\
\mathcal{F}^{-1}[\delta'(\omega + 2)] &= \frac{e^{-2it}}{2\pi i} t
\end{aligned}$$

(c)

$$\begin{aligned}
\mathcal{F}[\delta(t)] &= 1 \\
&\downarrow \\
\mathcal{F}^{-1}[2] &= 2\delta(t)
\end{aligned}$$

(d)

$$\begin{aligned}
\mathcal{F}[\theta(t+1) - \theta(t-1)] &= 2\frac{\sin(\omega)}{\omega} \\
\mathcal{F}\left[\theta\left(\frac{t}{2} + 1\right) - \theta\left(\frac{t}{2} - 1\right)\right] &= \mathcal{F}[\theta(t+2) - \theta(t-2)] = 2\frac{\sin(2\omega)}{\omega} \\
\mathcal{F}[(\theta(t+2) - \theta(t-2))'] &= \mathcal{F}[\delta(t+2) - \delta(t-2)] = i\omega\left(2\frac{\sin(2\omega)}{\omega}\right) = 2i\sin(2\omega) \\
&\downarrow \\
\mathcal{F}^{-1}[\sin(2\omega)] &= \frac{1}{2i}(\delta(t+2) - \delta(t-2))
\end{aligned}$$

(e)

$$\mathcal{F} [\delta'(t)] = i\omega$$

$$\mathcal{F} [-\delta''(t)] = \omega^2$$

$$\mathcal{F} [\delta(t)] = 1$$

$$\downarrow$$

$$\mathcal{F}^{-1} [\omega^2 + 2\omega + 1] = -\delta''(t) - 2i\delta'(t) + \delta(t)$$

□

D.34

Solution:

□

D.36

- (a) $\delta(t-1)$
- (b) $\delta'(t+2)$
- (c) 2
- (d) $\sin(2t)$
- (e) $(t+1)^2$
- (f) $e^t\theta(t)$

Solution:

$$\mathcal{L} [\delta(t)] = 1$$

$$\mathcal{L} [f(t-t_0)] = e^{-t_0s} \mathcal{L} [f(t)]$$

$$\mathcal{L} [f'(t)] = s\mathcal{L} [f(t)]$$

$$\mathcal{L} [e^{at}f(t)] = \mathcal{L} [f(t-a)]$$

$$\mathcal{L} [\theta(t)] = \frac{1}{s}$$

(a)

$$\mathcal{L} [\delta(t-1)] = e^{-s}$$

(b)

$$\mathcal{L} [\delta'(t+1)] = se^{2s}$$

(c) Not possible

(d) Not possible

(e) Not possible

$$\mathcal{L} [(t+1)^2] = \mathcal{L} [t^2 + 2t + 1]$$

(f)

$$\mathcal{L} [e^t \theta(t)] = \frac{1}{s-1}, \quad \operatorname{Re} s > 0$$

□

D.41

Expand the differential equations to $\mathbb{R}$ so that the boundary conditions are fulfilled.

(a)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u(0) = 1 \end{cases}$$

(b)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u'(0) = 1 \end{cases}$$

Solution:

Dirichlet conditions give odd expansion, Neumann conditions give even expansions.

(a)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u(0) = 1 \end{cases} \rightarrow (u^-)'' - iu^- = 2\delta', \quad x \in \mathbb{R}$$

(b)

$$\begin{cases} u''(x) - iu(x) = 0, & x > 0 \\ u'(0) = 1 \end{cases} \rightarrow (u^+)'' - iu^+ = 2\delta, \quad x \in \mathbb{R}$$

□

Hilbert Spaces

H.1

Solution:

☐

H.2

Solution:

☐

H.3

Solution:

☐

H.4

Solution:

☐

H.5a**Solution:**☐**H.11****Solution:**☐**H.12****Solution:**☐**H.13****Solution:**☐**H.15**

$$(u \mid v) = \int_{-\infty}^{\infty} u(x)v(x) \frac{e^{-x^2}}{\sqrt{\pi}} dx$$

Determine the first three Hermite polynomials

Solution:

Using Gram-Schmidt with $u_n = [1, x, x^2]$

$$\begin{aligned}\varphi_0 &= 1 \\ (\varphi_0 | \varphi_0) &= \int_{-\infty}^{\infty} \frac{e^{-x^2}}{\sqrt{\pi}} dx = 1 \\ \psi_0 &= \frac{\varphi_0}{(\varphi_0 | \varphi_0)} = 1\end{aligned}$$

Next

$$\begin{aligned}\varphi_1 &= u_1 - P_{[\varphi_0]} u_1 = u_1 - \frac{(\varphi_0 | u_1)}{(\varphi_0 | \varphi_0)} \varphi_0 = x - \frac{(1 | x)}{(1 | 1)} 1 = \\ &\quad \underbrace{x - \int_{-\infty}^{\infty} x \frac{e^{-x^2}}{\sqrt{\pi}} dx}_{=0 \text{ (odd function)}} = x \\ \psi_1 &= \frac{\varphi_1}{(\varphi_1 | \varphi_1)} = \frac{x}{(x | x)} \\ (x | x) &= \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2}}{\sqrt{\pi}} dx = -\frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} (x) (-2xe^{-x^2}) dx = \\ &= -\frac{1}{2\sqrt{\pi}} \left([xe^{-x^2}]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} e^{-x^2} dx \right) = \frac{1}{2}\end{aligned}$$

Finally this gives

$$\psi_1 = \frac{x}{(x | x)^{\frac{1}{2}}} = \sqrt{2}x$$

Next

$$\begin{aligned}
\varphi_2 &= u_2 - P_{[\varphi_0, \varphi_1]} u_2 = x^2 - \frac{(1 \mid x^2)}{(1 \mid 1)} 1 - \frac{(x \mid x^2)}{(x \mid x)} x = \\
&= x^2 - \int_{-\infty}^{\infty} x^2 \frac{e^{-x^2}}{\sqrt{\pi}} dx - \underbrace{\int_{-\infty}^{\infty} x^3 \frac{e^{-x^2}}{\sqrt{\pi}} dx}_{=0 \text{ odd function}} = x^2 - \frac{1}{2} \\
\psi_2 &= \frac{\varphi_2}{(\varphi_2 \mid \varphi_2)^{\frac{1}{2}}} = \left(x^2 - \frac{1}{2}\right) \left(\int_{-\infty}^{\infty} \left(x^2 - \frac{1}{2}\right)^2 \frac{e^{-x^2}}{\sqrt{\pi}} dx \right)^{-\frac{1}{2}} = \\
&= \left(x^2 - \frac{1}{2}\right) \left(\int_{-\infty}^{\infty} \left(x^4 - x^2 + \frac{1}{4}\right) \frac{e^{-x^2}}{\sqrt{\pi}} dx \right)^{-\frac{1}{2}} \\
&\quad \text{(i) } \int_{-\infty}^{\infty} \frac{1}{4} \frac{e^{-x^2}}{\sqrt{\pi}} dx = \boxed{\frac{1}{4}} \\
&\quad \text{(ii) } \int_{-\infty}^{\infty} -x^2 \frac{e^{-x^2}}{\sqrt{\pi}} dx = \boxed{-\frac{1}{2}} \\
&\quad \text{(iii) } \int_{-\infty}^{\infty} x^4 \frac{e^{-x^2}}{\sqrt{\pi}} dx = -\frac{1}{2\sqrt{\pi}} \int_{-\infty}^{\infty} (x^3)(-2xe^{-x^2}) dx = \\
&\quad = \underbrace{\left[x^2 e^{-x^2}\right]_{-\infty}^{\infty}}_{=0} - \underbrace{\int_{-\infty}^{\infty} 3x^2 e^{-x^2} dx}_{=\frac{3\sqrt{\pi}}{2}} = \boxed{\frac{3}{4}}
\end{aligned}$$

So

$$\psi_2 = \left(x^2 - \frac{1}{2}\right) \left(\frac{1}{4} - \frac{1}{2} + \frac{3}{4}\right)^{-\frac{1}{2}} = \left(x^2 - \frac{1}{2}\right) \sqrt{2}$$

The three first ortonormal Hermite polynomials are

$$\psi_0 = \boxed{1}$$

$$\psi_1 = \boxed{\sqrt{2}x}$$

$$\psi_2 = \boxed{\sqrt{2}\left(x^2 - \frac{1}{2}\right)}$$

□

H.18

$$\int_0^\pi (x - a \sin(x) - b \sin(2x))^2 dx$$

is minimized.

Solution:

$$v \in \mathcal{M} = [\sin(x), \sin(2x)]$$

$\sin(x)$ and $\sin(2x)$ are orthogonal and we can use the projection formula which says that the minimum is obtained when

$$v = P_{[\mathcal{M}]} u$$

$$v_1 = a \sin(x) = P_{[\sin(x)]} u = \frac{(\sin(x) | x)}{(\sin(x) | \sin(x))} \sin(x)$$

$$v_2 = b \sin(2x)$$

From this, a is calculated as

$$a = \frac{(\sin(x) | x)}{(\sin(x) | \sin(x))}$$

$$(\sin(x) | x) = \int_0^\pi x \sin(x) dx = -[x \cos(x)]_0^\pi + \int_0^\pi \pi \cos(x) dx = \pi$$

$$(\sin(x) | \sin(x)) = \int_0^\pi \sin^2 x dx = \int_0^\pi \frac{1 - \cos(2x)}{2} dx = \frac{\pi}{2}$$

↓

$$a = \frac{\pi}{\frac{\pi}{2}} = 2$$

and b is calculated in the same way as

$$b = \frac{(\sin(2x) | x)}{(\sin(2x) | \sin(2x))}$$

$$(\sin(2x) | x) = \int_0^\pi x \sin(2x) dx = -\left[\frac{x \cos(2x)}{2}\right]_0^\pi + \int_0^\pi \frac{\cos(2x)}{2} dx = -\frac{\pi}{2}$$

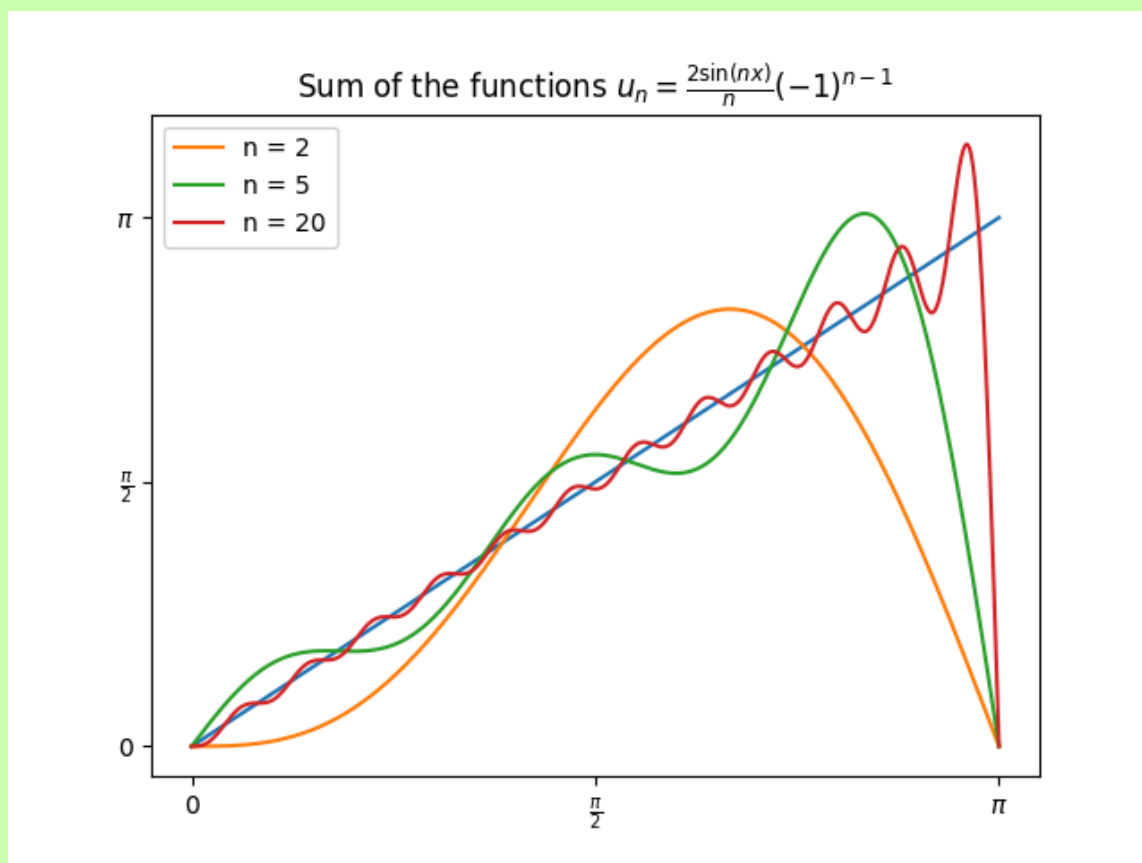
$$(\sin(2x) | \sin(2x)) = \int_0^\pi \sin^2(2x) dx = \int_0^\pi \frac{1 - \cos(2x)}{2} dx = \frac{\pi}{2}$$

↓

$$b = -\frac{\frac{\pi}{2}}{\frac{\pi}{2}} = -1$$

In general, the functions $v \in \mathcal{M} = \{\sin(nx)\}_{n=1}^{\infty}$ that best approximate $u(x) = x$ at the interval $[0, \pi]$ is given by

$$u_n = \frac{2(-1)^{n-1} \sin nx}{n}$$



□

H.19

(a) Give an orthogonal base in $\mathcal{V}$ with the scalar product

$$(u \mid v) = \int_{-1}^1 u(x)v(x) \, dx$$

(b) Let $u(x) = x^2 - x$. Determine $v \in \mathcal{V}$ so that

$$\int_{-1}^1 |u(x) - v(x)|^2 \, dx$$

is minimized.

(c) Determine the lowest possible value of the integral in (b).

Solution:(a) $\square$ **H.20****Solution:**

For this setup, the scalar product is defined as

$$(u | v) = \int_0^1 u(x)v(x)w(x) \, dx = \int_0^1 u(x)v(x)x \, dx$$

Find an orthogonal base

$$\varphi_0 = 1$$

$$(\varphi_0 | \varphi_0) = \int_0^1 x \, dx = \frac{1}{2}$$

$$\psi_0 = \frac{\varphi_0}{(\varphi_0 | \varphi_0)} = 2$$

$$\varphi_1 = x - P_{[\varphi_0]}x = x - \frac{(\varphi_0 | x)}{(\varphi_0 | \varphi_0)}\varphi_0 = x - 2 \int_0^1 x^2 \, dx = x - \frac{2}{3}$$

$$(\varphi_1 | \varphi_1) = \int_0^1 \left(x - \frac{2}{3}\right)^2 x \, dx = \frac{1}{36}$$

$$\psi_1 = \frac{\varphi_1}{(\varphi_1 | \varphi_1)} = 36x - 24$$

Now $v \in \mathcal{M} = \{2, 36x - 24\}$

Are looking for

$$\int_0^1 (x^4 - a\psi_0 - b\psi_1)^2 x \, dx$$

From which we can calculate

$$a\psi_0 = \frac{(\psi_0 | x^4)}{(\psi_0 | \psi_0)}\psi_0$$

$$a = \frac{(\psi_0 | u)}{(\psi_0 | \psi_0)}$$

$$(\psi_0 | u) = \int_0^1 2x^5 \, dx = \frac{1}{3}$$

$$(\psi_0 | \psi_0) = \int_0^1 4x \, dx = 2$$

$$a = \frac{1}{6}$$

and

$$b\psi_1 = \frac{(\psi_1 | x^4)}{(\psi_1 | \psi_1)}\psi_1$$

$$b = \frac{(\psi_1 | x^4)}{(\psi_1 | \psi_1)}$$

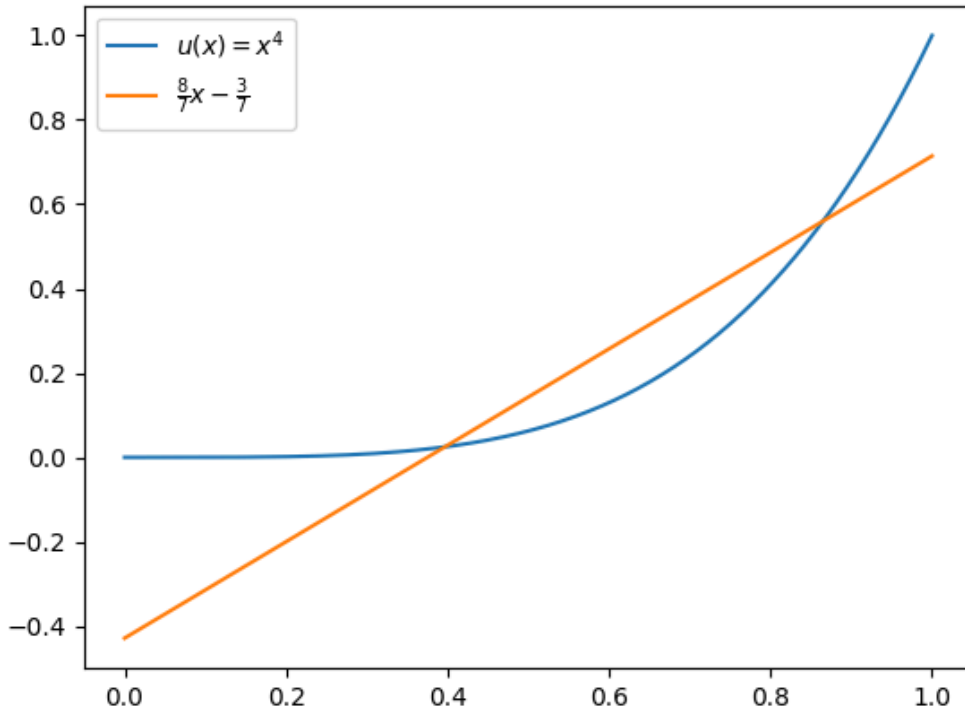
$$(\psi_1 | x^4) = \int_0^1 (36x^6 - 24x^5) \, dx = \frac{36}{7} - \frac{24}{6} = \frac{8}{7}$$

$$(\psi_1 | \psi_1) = \int_0^1 (1296x^3 - 1728x^2 + 576x) \, dx = 36$$

$$b = \frac{2}{63}$$

The first degree polynomial that best approximates the function $u(x) = x^4$ in $I = [0, 1]$ is

$$\frac{1}{6}\psi_0 + \frac{2}{63}\psi_1 = \frac{8}{7}x - \frac{3}{7}$$



(b) We are now looking for a second degree polynomial which means our $\psi \in \mathcal{M} = \{1, x, x^2\}$ ψ_0 and ψ_1 were already calculated in (a) so

$$\varphi_2 = u_2 - P_{[\varphi_0, \varphi_1]} u_2 = x^2 - \frac{(\varphi_0 | x^2)}{(\varphi_0 | \varphi_0)} \varphi_0 - \frac{(\varphi_1 | x^2)}{(\varphi_1 | \varphi_1)} \varphi_1$$

$$\frac{(\varphi_0 | x^2)}{(\varphi_0 | \varphi_0)} \varphi_0 = 2(\varphi_0 | x^2) = 2 \int_0^1 x^3 dx = \frac{1}{2}$$

$$\frac{(\varphi_1 | x^2)}{(\varphi_1 | \varphi_1)} \varphi_1 = (\varphi_1 | x^2)(36x - 24)$$

Now calculating

$$(\varphi_1 | x^2) = \int_0^1 \left(x^4 - \frac{2x^3}{3} \right) dx = \frac{1}{30}$$

which gives finally

$$\varphi_2 = x^2 - \frac{6}{5}x + \frac{3}{10}$$

$$(\varphi_2 | \varphi_2) = \int_0^1 \left(x^2 - \frac{6x}{5} + \frac{3}{10} \right)^2 x \, dx = \frac{1}{600}$$

$$\psi_2 = 600x^2 - 720x + 180$$

We can now find the best approximation with

$$\int_0^1 (x^4 - a\psi_0 - b\psi_1 - c\psi_2)^2 x \, dx$$

$$c\psi_2 = \frac{(\psi_2 | x^4)}{(\psi_2 | \psi_2)} \psi_2$$

$$c = \frac{(\psi_2 | x^4)}{(\psi_2 | \psi_2)}$$

$$(\psi_2 | x^4) = \int_0^1 (600x^2 - 720x + 180)x^4 \, dx = \frac{15}{7}$$

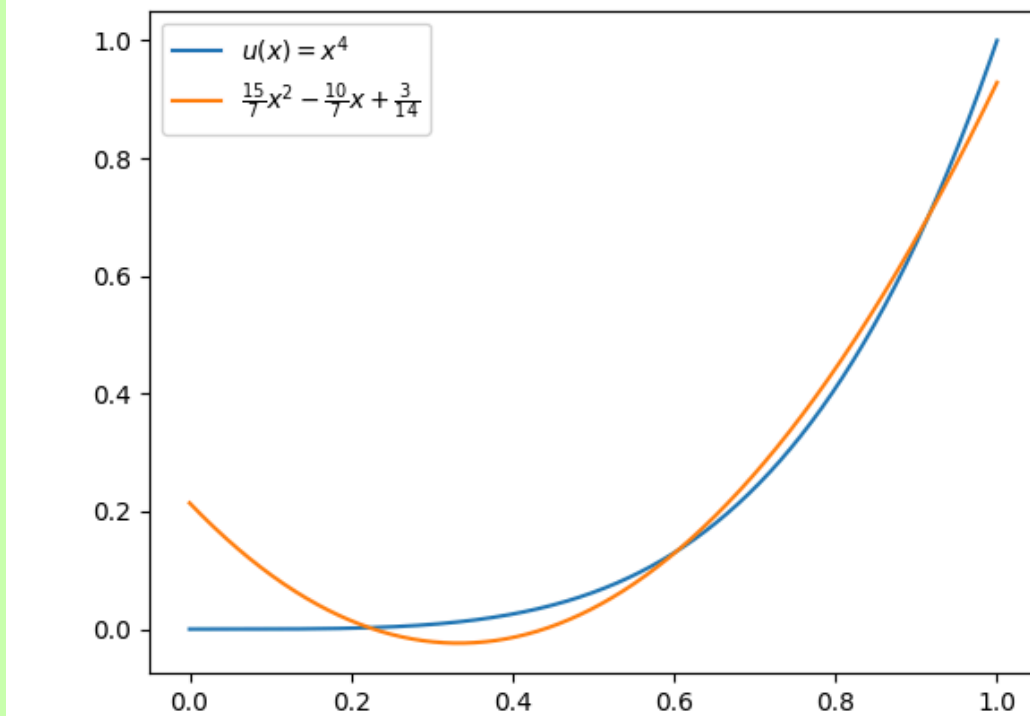
$$(\psi_2 | \psi_2) = \int_0^1 (600x^2 - 720x + 180)^2 x \, dx = 600$$

$$\downarrow$$

$$c = \frac{15}{4200} = \frac{1}{280}$$

The best second degree polynomial to approximate $u(x) = x^4$ is then

$$a\psi_0 + b\psi_1 + c\psi_2 = \frac{15}{7}x^2 - \frac{10}{7}x + \frac{3}{14}$$



□

H.27

- (a) Show that p_n is orthogonal to all polynomials of degree $p < n$
- (b) Show that if p is a polynomial, not identically zero, such that $(p_k | p) = 0$ for $k = 0, 1, 2$, then degree of $p \geq 3$.

Solution:

- (a) All polynomials of degree m can be written

$$p = \sum_{k=0}^m c_k p_k$$

Gives

$$(p_n | p) = \left(p_n \left| \sum_{k=0}^m c_k p_k \right. \right) = \sum_{k=0}^m c_k (p_n | p_k) = 0 \text{ for } m < n$$

since all polynomials are pairwise orthogonal

(b)

$$p = \sum_{k=0}^m c_k p_k$$

where

$$c_k = \frac{(p_k | p)}{(p_k | p_k)}$$

$$(p_k | p) = 0, \text{ for } k = 0, 1, 2 \rightarrow c_0 = c_1 = c_2 = 0$$

The restriction that $p \neq 0$ gives that $\deg p \geq 3$ $\square$

H.30

Solution:

$\square$

H.31

Solution:

$\square$

H.33

$$f_n(x) = x^n, \quad x \in I = [0, 1], \quad n = 0, 1, \dots$$

Determine $\|f_n\|_p$ for $p = 1, 2$ and ∞ . Is it for any of these norms true that $f_n \rightarrow 0$ when $n \rightarrow \infty$? What is the pointwise limit function?

Solution:

$$\|f_n\|_1 = \int_0^1 |x^n| \, dx = \left[\frac{x^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1} \rightarrow 0, \quad n \rightarrow \infty$$

$$\|f_n\|_2 = \left(\int_0^1 |x^n|^2 \, dx \right)^{\frac{1}{2}} = \left(\left[\frac{x^{2n+1}}{2n+1} \right]_0^1 \right)^{\frac{1}{2}} = \frac{1}{\sqrt{2n+1}} \rightarrow 0, \quad n \rightarrow \infty$$

$$\|f_n\|_\infty = \sup_{x \in [0,1]} |f_n| = \sup_{x \in [0,1]} |x^n| = 1 \rightarrow 1, \quad n \rightarrow \infty$$

□

H.36

Solution:

□

H.38

Solution:

□

H.39

Solution:

□

H.45**Solution:**

□

H.47**Solution:**

□

H.50

$$\mathcal{A} = -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} \right)$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(-1, 1) \mid u \text{ and } u' \text{ bounded/limited}\}$$

- (a) Determine a scalar product in which $\mathcal{A}$ is symmetric.
 (b) Show that $\mathcal{A}$ is symmetric in the scalar product from (a), using the definition.
 (c) Show that $1, x, 2x^2 - 1$ are eigenfunctions to $\mathcal{A}$ and determine the corresponding eigenvalues.
 (d) Evaluate

$$\int_{-1}^1 \frac{2x^2 - 1}{\sqrt{1-x^2}} dx$$

Solution:

- (a) The operator is a Sturm-Liouville operator

$$\mathcal{A}u = -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} \right) = -\frac{1}{w} (pu')', \quad w(x) = \frac{1}{\sqrt{1-x^2}}$$

and the scalar product is then

$$(u \mid v) = \int_{-1}^1 \overline{u(x)} v(x) \frac{1}{w(x)} dx$$

- (b) Using the scalar product definition

$$(u \mid v) = \int \overline{u(x)} v(x) w(x) \, dx, \quad w(x) = \frac{1}{\sqrt{1-x^2}}$$

we get

$$\begin{aligned}
 (u \mid \mathcal{A}v) &= \\
 &= \int_{-1}^1 \overline{u(x)} [\mathcal{A}v(x)] \frac{1}{w(x)} \, dx = \int_{-1}^1 \overline{u(x)} \left[-\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} v(x) \right) \right] \frac{1}{\sqrt{1-x^2}} \, dx = \\
 &= - \int_{-1}^1 \overline{u(x)} \left(\sqrt{1-x^2} v'(x) \right)' \, dx = - \underbrace{\left[\overline{u(x)} \sqrt{1-x^2} v'(x) \right]_{-1}^1}_{=0} + \int_{-1}^1 \overline{u'(x)} \sqrt{1-x^2} v'(x) \, dx = \\
 &= \int_{-1}^1 \left(\overline{u'(x)} \sqrt{1-x^2} \right) v'(x) \, dx = \underbrace{\left[\left(\overline{u'(x)} \sqrt{1-x^2} \right) v(x) \right]_{-1}^1}_{=0} - \int_{-1}^1 \left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \, dx = \\
 &= \int_{-1}^1 - \left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \, dx = \int_{-1}^1 -\sqrt{1-x^2} \left(\overline{u'(x)} \sqrt{1-x^2} \right)' v(x) \frac{1}{\sqrt{1-x^2}} \, dx = \\
 &= \int_{-1}^1 \left[-\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} \overline{u(x)} \right) \right] v(x) \frac{1}{\sqrt{1-x^2}} \, dx = \\
 &= \int_{-1}^1 [\mathcal{A}u(x)] v(x) \frac{1}{w(x)} \, dx = (\mathcal{A}u \mid v)
 \end{aligned}$$

(c)

$$\mathcal{A}u = \lambda u$$

$$u(x) = 1$$

$$\downarrow$$

$$\mathcal{A} \cdot 1 = \lambda_0 \cdot 1 \rightarrow -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} \cdot 1 \right) = \lambda_0 \cdot 1 \rightarrow 0 = \lambda_0 \cdot 1 \rightarrow \boxed{\lambda_0 = 0}$$

$$u(x) = x$$

$$\downarrow$$

$$\mathcal{A}x = \lambda_1 x \rightarrow -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} x \right) = \lambda_1 x \rightarrow -\sqrt{1-x^2} \frac{-2x}{2\sqrt{1-x^2}} = \lambda_1 x \rightarrow$$

$$\rightarrow x = \lambda_1 x \rightarrow \boxed{\lambda_1 = 1}$$

$$u(x) = 2x^2 - 1$$

$$\downarrow$$

$$\begin{aligned} \mathcal{A}(2x^2 - 1) &= \lambda_2(2x^2 - 1) \rightarrow -\sqrt{1-x^2} \frac{d}{dx} \left(\sqrt{1-x^2} \frac{d}{dx} (2x^2 - 1) \right) = \lambda_2(2x^2 - 1) \rightarrow \\ &\rightarrow -\sqrt{1-x^2} \frac{d}{dx} (4x\sqrt{1-x^2}) = -\sqrt{1-x^2} \left(4\sqrt{1-x^2} - \frac{8x^2}{2\sqrt{1-x^2}} \right) = -4 + 4x^2 + 4x^2 = \\ &= 8x^2 - 4 = \lambda_2(2x^2 - 1) \rightarrow 4(2x^2 - 1) = \lambda_2(2x^2 - 1) \rightarrow \boxed{\lambda_2 = 4} \end{aligned}$$

(d) The integral can be rewritten with the scalar product in (a) as

$$\int_{-1}^1 \frac{2x^2 - 1}{\sqrt{1-x^2}} dx = (1 \mid 2x^2 - 1), \quad w(x) = \frac{q}{\sqrt{1-x^2}}$$

Since the functions 1 and $2x^2 - 1$ are eigenfunctions, the scalar product between them are pairwise orthogonal and therefore = 0.

□

H.51

Solution:

□

H.52**Solution:**

□

Operator

$$\mathcal{A} = -\text{laplace} = -\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}$$

Solution:

$$\begin{aligned}
(u \mid \mathcal{A}v) &= \int_0^a \int_0^b \bar{u}(-v''_{xx} - v''_{yy}) \, dx \, dy = \\
(1) &= - \int_0^b \underbrace{\left(\int_0^a \bar{u} v''_{xx} \, dx \right)}_{(i-x)} \, dy - \int_0^a \underbrace{\left(\int_0^b \bar{u} v''_{yy} \, dy \right)}_{(i-y)} \, dx \\
(i-x) &= \underbrace{[\bar{u} v'_x]_0^a}_{=0} - \int_0^a \bar{u}'_x v'_x \, dx \\
(i-y) &= \underbrace{[\bar{u} v'_y]_0^b}_{=0} - \int_0^b \bar{u}'_y v'_y \, dy \\
&\downarrow \\
(1) &= \int_0^b \underbrace{\left(\int_0^a \bar{u}'_x v'_x \, dx \right)}_{(ii-x)} \, dy + \int_0^a \underbrace{\left(\int_0^b \bar{u}'_y v'_y \, dy \right)}_{(ii-y)} \, dx = (***) \\
(ii-x) &= \underbrace{[\bar{u}'_x v'_x]_0^a}_{=0} - \int_0^a \bar{u}''_{xx} v \, dx \\
(ii-y) &= \underbrace{[\bar{u}'_y v'_y]_0^b}_{=0} - \int_0^b \bar{u}''_{yy} v \, dy \\
&\downarrow \\
(1) &= - \int_0^b \int_0^a \bar{u}''_{xx} v \, dx \, dy - \int_0^a \int_0^b \bar{u}''_{yy} v \, dy \, dx = \int_0^a \int_0^b (-\bar{u}''_{xx} - \bar{u}''_{yy}) v \, dx \, dy = \\
&= (\mathcal{A}u \mid v)
\end{aligned}$$

To show positive semidefinite, use the form described on (***) and $u = v$:

$$(***) = \int_0^a \int_0^b (\bar{u}'_x u'_x + \bar{u}'_y u'_y) \, dx \, dy = \int_0^a \int_0^b (|u'_x|^2 + |u'_y|^2) \, dx \, dy \geq 0$$

To find the eigenvalues associated with the operator, use separation of variables

$$\begin{aligned}
u(x, y) &= X(x)Y(y) \\
\mathcal{A}u &= \lambda u \rightarrow -X''Y - XY'' = \lambda XY \rightarrow -\frac{X''}{X} - \frac{Y''}{Y} = \lambda \\
&\downarrow \\
&\begin{cases} \frac{X''}{X} = -\gamma_x^2, & X(0) = X(a) = 0 \\ \frac{Y''}{Y} = -\gamma_y^2, & Y'(0) = Y'(b) = 0 \end{cases}
\end{aligned}$$

Starting with $X(x)$:

$$X'' = -\gamma_x^2 X \rightarrow X(x) = e^{i\gamma_x x} = A \cos(\gamma_x x) + B \sin(\gamma_x x)$$

$$X(0) = A = 0 \rightarrow A = 0$$

$$X(a) = B \sin(\gamma_x a) = 0 \rightarrow \gamma_x = \frac{\pi k_x}{a}, k_x = 1, 2, 3 \dots$$

↓

$$X(x) = B \sin\left(\frac{\pi k_x x}{a}\right) \quad \square$$

Now $Y(y)$:

$$Y'' = -\gamma_y^2 Y \rightarrow Y(y) = e^{i\gamma_y y} = A \cos(\gamma_y y) + B \sin(\gamma_y y)$$

$$Y'(x) = -\gamma_y A \sin(\gamma_y y) + \gamma_y B \cos(\gamma_y y)$$

$$Y'(0) = \gamma_y B = 0 \rightarrow B = 0$$

$$Y'(b) = -\gamma_y A \sin(\gamma_y b) = 0 \rightarrow \gamma_y = \frac{\pi k_y}{b}, k_y = 0, 1, 2 \dots$$

↓

$$Y(y) = A \cos\left(\frac{\pi k_y y}{b}\right) \quad \square$$

□

H.8 Operators in Hilbert Spaces

Linear avbildning

$$\mathcal{A} : \mathcal{H}_1 \rightarrow \mathcal{H}_2$$

where $\mathcal{H}_1$ and $\mathcal{H}_2$ are two pre-Hilbert Spaces. Linearity:

$$\mathcal{A}(\lambda u + \mu v) = \lambda \mathcal{A}u + \mu \mathcal{A}v$$

is termed a **linear operator**.

Example H.19

Let I be a compact interval on $\mathbb{R}$, and let $K(x, y)$ be continuous for $x, y \in I$.

$$v(x) = \int_I K(x, y) u(y) \, dy$$

This defines an **integral operator**

$$\mathcal{K} : L_2(I) \rightarrow L_2(I)$$

The function $K(x, y)$ is termed the operators **kernel**

Special Functions

S.1

Solution:

Using the definition of $\Gamma(z)$:

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

it can be shown that $\Gamma(z+1) = z\Gamma(z)$:

$$\begin{aligned} \Gamma(z+1) &= \int_0^{\infty} t^z e^{-t} dt = \underbrace{[-t^z e^{-t}]_0^{\infty}}_{=0} + \int_0^{\infty} z t^{z-1} e^{-t} dt = \\ &= z \int_0^{\infty} t^{z-1} e^{-t} dt = z\Gamma(z) \end{aligned}$$

We also have

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = [-e^{-t}]_0^{\infty} = 1$$

This gives

$$\Gamma(3) = \Gamma(2+1) = 2\Gamma(2) = 2\Gamma(1+1) = 2 \cdot 1 \cdot \Gamma(1) \rightarrow \Gamma(3) = 2$$

Next

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt \rightarrow [t = x^2 \rightarrow dt = 2x dx] = 2 \int_0^{\infty} x x^{-1} e^{-x^2} dx = \\ &= 2 \int_0^{\infty} e^{-x^2} dx = 2 \frac{\sqrt{\pi}}{2} \rightarrow \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \end{aligned}$$

Finally we will use the definition for $\Gamma(z)$, $\text{Re } z \leq 0$:

$$\Gamma(z) = \frac{\Gamma(z+n)}{(z+n-1)\dots(z+1)z}, \quad n \text{ chosen such that } \operatorname{Re}(z+n) > 0$$

This gives

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}} = \frac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}} \rightarrow \Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}$$

□

S.5

Solution:

□

S.6

Solution:

□

S.7

$$\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

Solution:

Using the Fourier expansion

$$e^{ir \sin \theta} = \sum_{-\infty}^{\infty} J_n(r) e^{in\theta}$$

and Parseval's Theorem

$$\begin{aligned}
\frac{1}{2\pi} \int_{-\pi}^{\pi} \overline{e^{ir \sin \theta}} e^{ir \sin \theta} d\theta &= \sum_{-\infty}^{\infty} \overline{J_n(r)} J_n(r) e^{in\theta} e^{in\theta} \\
&\downarrow \\
\frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta &= \sum_{-\infty}^{\infty} |J_n(r)|^2 \\
&\downarrow \\
\frac{1}{2\pi} 2\pi &= \sum_{-\infty}^{\infty} |J_n(r)|^2 = 1
\end{aligned}$$

□

S.11

$$\frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) = -\frac{J_{\nu+1}(x)}{x^\nu}$$

(b) Use (a) and

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

to calculate $J_{\frac{3}{2}}(x)$

(c) Show that $J'_0(x) = -J_1(x)$

Solution:

The power series is

$$\begin{aligned}
J_\nu(x) &= \left(\frac{x}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k \\
&\downarrow \\
\frac{J_\nu(x)}{x^\nu} &= \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k
\end{aligned}$$

This can be differentiated termwise as

$$\begin{aligned}
\frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) &= \left(\frac{1}{2} \right)^\nu \sum_{k=0}^{\infty} \frac{-2xk}{4k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4} \right)^{k-1} = \\
&= -\frac{x}{2} \left(\frac{1}{2} \right)^\nu \sum_{\substack{k=1 \\ *}}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4} \right)^{k-1} = \\
&= -\frac{1}{x^\nu} \left(\frac{x}{2} \right)^{\nu+1} \sum_{k=1}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4} \right)^{k-1} \rightarrow [k \rightarrow k+1] \rightarrow \\
&\rightarrow -\frac{1}{x^\nu} \left(\frac{x}{2} \right)^{\nu+1} \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + 1 + k + 1)} \left(-\frac{x^2}{4} \right)^k = \boxed{-\frac{J_{\nu+1}(x)}{x^\nu}}
\end{aligned}$$

* (The $k=0$ contribution is zero, since it is a constant and was differentiated.)

(b) Setting $\nu = \frac{1}{2}$ we get

$$\begin{aligned}
\frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) &= -\frac{J_{\frac{3}{2}}(x)}{x^{\frac{1}{2}}} \\
\frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) &= \frac{d}{dx} \left(\frac{\sqrt{\frac{2}{\pi x}} \sin x}{x^{\frac{1}{2}}} \right) = \frac{d}{dx} \left(\sqrt{\frac{2}{\pi}} \frac{\sin x}{x} \right) = \sqrt{\frac{2}{\pi}} \left(-\frac{1}{x^2} \sin x + \frac{1}{x} \cos x \right) \\
&\downarrow \\
J_{\frac{3}{2}}(x) &= x^{\frac{1}{2}} \sqrt{\frac{2}{\pi}} \left(\frac{\sin x}{x^2} - \frac{x}{x^2} \cos x \right) = \sqrt{\frac{2}{\pi}} x^{\frac{3}{2}} \left(\frac{\sin x - x \cos x}{x^3} \right)
\end{aligned}$$

(c) Using the result from (a)

$$\frac{d}{dx} \left(\frac{J_0(x)}{x^0} \right) = -\frac{J_1(x)}{x^0} \rightarrow \boxed{J_{0'}(x) = -J_1(x)}$$

□

S.12

Solution:

□

S.13

(a) $r^2 y'' + r y' = 0$

(b) $r^2 y'' + r y' - y = 0$

(c) $r^2 y'' + r y' + r^2 y = 0$

Solution:

Using a definition that the general solution to Bessel's differential equations

$$u'' + \frac{1}{r}u' + \left(\lambda - \frac{\nu^2}{r^2}\right)u = 0$$

are

$$\begin{aligned} & aJ_\nu(\sqrt{\lambda}r) + bY_\nu(\sqrt{\lambda}r), & \text{if } \lambda > 0 \\ & ar^\nu + br^{-\nu}, & \text{if } \lambda = 0, \nu \neq 0 \\ & a + b \ln r, & \text{if } \lambda = \nu = 0 \end{aligned}$$

(a)

$$r^2 y'' + r y' = 0 \rightarrow y'' + \frac{1}{r}y' = 0, \quad \lambda = \nu = 0$$

$$y(r) = a + b \ln r$$

(b)

$$r^2 y'' + r y' - y = 0 \rightarrow y'' + \frac{1}{r}y' - \frac{1}{r^2}y = 0, \quad \lambda = 0, \nu = 1$$

$$y(r) = ar + \frac{b}{r}$$

(c)

$$r^2 y'' + r y' + r^2 y = 0 \rightarrow y'' + \frac{1}{r}y' + y = 0, \quad \lambda = 1, \nu = 0$$

$$y(r) = aJ_0(r) + bY_0(r)$$

□

S.14

$$\mathcal{A}u = -\frac{1}{x}(xu')'$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(I) \mid u \text{ bounded near } 0, u'(2) = 0\}$$

- (a) Determine a scalar product in which $\mathcal{A}$ is symmetric and positive semidefinite
- (b) Determine the eigenvalues and eigenvectors of $\mathcal{A}$.
- (c) Determine numerically the three lowest eigenvalues.
- (d) Let $u(x) = x$. Write down u 's expansion in eigenfunctions to $\mathcal{A}$. The coefficients should be given in integral form.

Solution:

(a)

$$(u \mid \mathcal{A}v) = \int_0^2 \overline{u(x)} \mathcal{A}v(x) w(x) dx$$

with $w(x) = x$

$$\begin{aligned} (u \mid \mathcal{A}v) &= - \int_0^2 \overline{u} (xv')' dx = - \underbrace{[\overline{u}(xv')]_0^2}_{=0} + \int_0^2 (\overline{u}'x) v' dx = \\ &= \underbrace{\int_0^2 (x\overline{u}') v' dx}_{= (*)} = \underbrace{[(x\overline{u}')v]_0^2}_{=0} - \int_0^2 (x\overline{u}')' v dx = (\mathcal{A}u \mid v) \end{aligned}$$

To show that it's positive semidefinite, start from (*) and set $u = v$:

$$\int_0^2 x \overline{u}' u' dx = \int_0^2 x |u'|^2 dx \geq 0$$

(b) Setting

$$\begin{aligned} \mathcal{A}u = \lambda u &\rightarrow -\frac{1}{x}(xu')' = -\frac{1}{x}(u' + xu'') = \lambda u \\ &\downarrow \\ -\frac{1}{x}xu'' - \frac{1}{x}u' - \lambda u &= 0 \rightarrow u'' + \frac{1}{x}u' + \lambda u = 0 \end{aligned}$$

which has the solutions (according to the definition of the Bessel's differential equation solutions):

$$\begin{aligned} aJ_0(\sqrt{\lambda}x) + bY_0(\sqrt{\lambda}x), & \quad \lambda > 0, \nu = 0 \\ a + b \ln r, & \quad \lambda = \nu = 0 \end{aligned}$$

Since Y_0 is unbounded near $x = 0$, b has to be zero to fulfill the requirement of u being bounded near zero.

$$u(x) = aJ_0(\sqrt{\lambda}x)$$

$$u'(2) = \sqrt{\lambda}aJ_0'(\sqrt{\lambda}2) = 0$$

This gives $\lambda_0 = 0$. The rest of the eigenvalues are given by

$$J_0'(2\sqrt{\lambda_k}) = 0 \rightarrow \lambda_k = \left(\frac{\beta_{0,k}}{2}\right)^2$$

The eigenfunctions are then given by

$$\varphi_{k(x)} = J_0\left(\frac{\beta_{0,k}}{2}x\right)$$

(c)

$$\lambda_0 = 0$$

$$\lambda_1 = \left(\frac{3.8317}{2}\right)^2 = 3.67$$

$$\lambda_2 = \left(\frac{7.0156}{2}\right)^2 = 12.31$$

□

S.15

Solution:

□

S.16

Solution:

□

S.18

Solution:

☐

S.20

Solution:

☐
